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Microscopic investigation of pollen morphology of Brassicaceae from Central Punjab-Pakistan

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Abstract

The intention of the current study is to provide an account on the palynological features of Brassicaceae from Central Punjab-Pakistan as a basis for future studies. Different morpho-palynological characteristics both qualitative and quantitative were analyzed during this research which includes shape of pollen, diameter of pollen, P/E ratio, exine sculpturing, thickness of exine, type of pollen, shape and size of lumens, and thickness of murus. Taxonomic keys were also constructed based on pollen morphological characters for correct identification of species. This study aims to provide detailed information of pollen diversity and their exine structure based on both qualitative and quantitative characters by using Light microscopy and Scanning electron microscopy. Shape of pollen is mostly prolate, but some species also have sub-prolate to spheroidal prolate types. Exine ornamentation in most species was reticulate, whereas micro reticulate (one species) and coarsely reticulate (one species) exine also observed in some pollen. All the pollen mentioned in this study have tricolpate apertures. Variation found in thickness of exine and other characters proved to be helpful at generic and specific level. The results reinforced the significance of pollen morphological features of family Brassicaceae and aid for valuable taxonomic tool in plant systematics.

KEYWORDS

Brassicaceae, Central Punjab, light microscopy, Pakistan, pollen morphology, scanning electron microscopy

1 | INTRODUCTION

Brassicaceae is consider among the largest families of angiosperms as it comprises 350 genera and approximately 3,000 species. In Pakistan, the Brassicaceae has 92 genera and about 250 species which include only five genera of cultivated species. The family Brassicaceae is distributed mainly to Irano-Turanian, Mediterranean, and Saharo-Sindian regions (Hedge, Vaughan, Macleod, & Jones, 1976).

Previously many work has been performed on pollen morphology of family Brassicaceae, and all of them demonstrated the significance of pollen morphology to understand the taxonomy of the family (Anchev & Deneva, 1997; Khalik Abdel, Van Den Berg, Van Der Maesen, & El Hadidi, 2002; Perveen, Qaiser, & Khan, 2004). Pollen morphology also consider as a tool for delimitation among the genus and species of family Brassicaceae (Brochmann, 1992; Keshavarzi, Abassian, & Sheidai, 2012; Khan, 2004; Mutlu & Erik, 2012).

In Pakistan, many work was done previously by Perveen et al. (2004) related to pollen morphology of family Brassicaceae, Khan (2004) investigated pollen morphology of genus *Arabidopsis* of family Brassicaceae, Khan (2005) studied pollen morphology of two genus of family Brassicaceae, and genus *Alyssum* of Brassicaceae also been investigated in terms of pollen morphology (Khan, 2003). Still, there is a gap in pollen morphology of family Brassicaceae due to not much intension to the flora of arid lands of Central Punjab-Pakistan.

Arid land vegetation is the important part of world's flora (Ashfaq et al. 2018). Previously much work has been done on family Pontederiaceae, some work is also done on family Haloragaceae (Aiken, 1978; Cook, 1998), and some medicinal plants of family Mimosaceae has also been studied regarding palynology by Bhunia and Mondal (2012) who also worked on Nymphaeaceae. Pollen study of Lemnaceae family (Aiken, 1978; Landolt, 1998), family Potamogetonaceae (Sorsa, 1988), Typhaceae family (Cook, 1998), Podostemaceae (de Sá-Haiad et al., 2010), and Eriocaulaceae family (De borges, Dos, & Giulietti, 2009) were studied.

Pollen morphology of medicinal plants was done for the first time in Pakistan by Perveen and Qaiser (1999). Sardar et al. (2013) studied different aspects of plants including anatomy, morphology, palynology, and their medicinal uses. However, at present, there is no separate documentation on the pollen morphology of some medicinal species belonging to family Brassicaceae from the study site. The forgoing investigation indicate that there is no definitive agreement for which level of classification can be useful of using palynological features in Brassicaceae and no single reports are found for the pollen morphology of species of this family. The aim of this study is to provide detailed information of pollen diversity and their exine structure based on both qualitative and quantitative characters using Light microscopy (LM) and Scanning electron microscopy (SEM) as a basis for future studies. Therefore, the present study was made to distinguish micro-morphological characters to strengthen the identification of complicated species of the family. The intention of present study is to determine the pollen fertility of selected species of family Brassicaceae which are economically important, and this will help to identify the freely reproducing species for conservation purposes.

2 | MATERIAL AND METHODS

2.1 | Collection of seed samples

Plants belonging to family Brassicaceae were collected through extensive field trips which were carried out in different dry or wetland areas of Central Punjab-Pakistan. The plant species were collected during May 2018 to August 2019. After collection, the species were identified by comparing their morphological characters with the species mentioned in Flora of Pakistan (Ali, 1980). These identified species were compared with the herbarium samples present in Herbarium of Pakistan (ISL) in Quaid-i-Azam University Islamabad, Pakistan. Identified plant species were then dried and pressed using blotting papers; after drying, they were mounted on herbarium sheets and then submitted to Herbarium of Pakistan (ISL). The voucher numbers along with area, altitude are given in (Table 1).

2.2 | LM analysis of pollen grains

During field trip for pollen analysis, mature flowers were collected separately. From these flowers, the anthers were separated and placed on glass slide. Pour one drop of acetic acid on these anthers

TABLE 1 List of plant species and its collection sites along with voucher numbers

Sr. No.	Name of taxa	Collection site	Altitude (m)/(feet)	Flowering period	Habitat	Voucher number
1.	*Alliaria petiolata* (M.Bieb.) Cavara & Grande	Marala Headworks (Sialkot)	250/820	April–August	Dry places, near the bank of road	MHA-106
2.	*Brassica rapa* L.	Gujranwala	226/744	April–June	Dry side areas	MHA-126
3.	*Capsella bursa-pastoris* (L.) Medik.	Gujrat	239/784.12	March–June	Moist places	MHA-90
4.	*Cardamine hirsuta* L.	Marala Headworks (Sialkot)	250/820	March–May	Mist bands of canals	MHA-87
5.	*Lepidium didymum* L.	Gujranwala	226/744	March–June	Waste lands, moist places	MHA-114
6.	*Nasturtium microphyllum* Boenn. ex Reichenb	Pabbi Forest (Kharrian)	280 m	April–July	Dense forest or shady place	MHA-123
7.	*Nasturtium officinale* W.T.Aiton	Chenab river (Gujrat)	829 ft	April–July	Moist and dry places	MHA-99
8.	*Rorippa indica* (L.) Hiern	Gujranwala	226/744	April–June	Damp places	MHA-104
9.	*Rorippa islandica* (Oeder) Borbás	Chicherwali canal (Gujranwala)	226/744	April–July	Wet areas	MHA-121
10.	*Sisymbrium irio* L.	Chicherwali canal (Gujranwala)	226/744	March–June	Road side areas of canal	MHA-88

and crushed them with the help of glass rod. After crushing, remove the debris with the help of camel hair brush and stained the slide with glycerin jelly. Mean values were calculated by taking 10 readings of each pollen character; polar diameter, equatorial diameter, exine thickness, length and width of colpi, lumen, and murus thickness. Meiji Techno Japan microscope (Model: MT4300H) were used to study both qualitative and quantitative characteristics of pollen. Pollen micrographs were taken on 40× resolutions of Leica Dialux 20 light microscope. Shape of the pollen grains was determined by P.E ratio. For the terminologies of pollens, the updated terminology of Punt, Hoen, Blackmore, Nilsson, and Le thomas (2007) and Grant-Downton (2009) was followed. For statistical analysis (SPSS 16.0), software was used to calculate minimum, maximum, mean, and standard error of each quantitative feature (Plate 1).

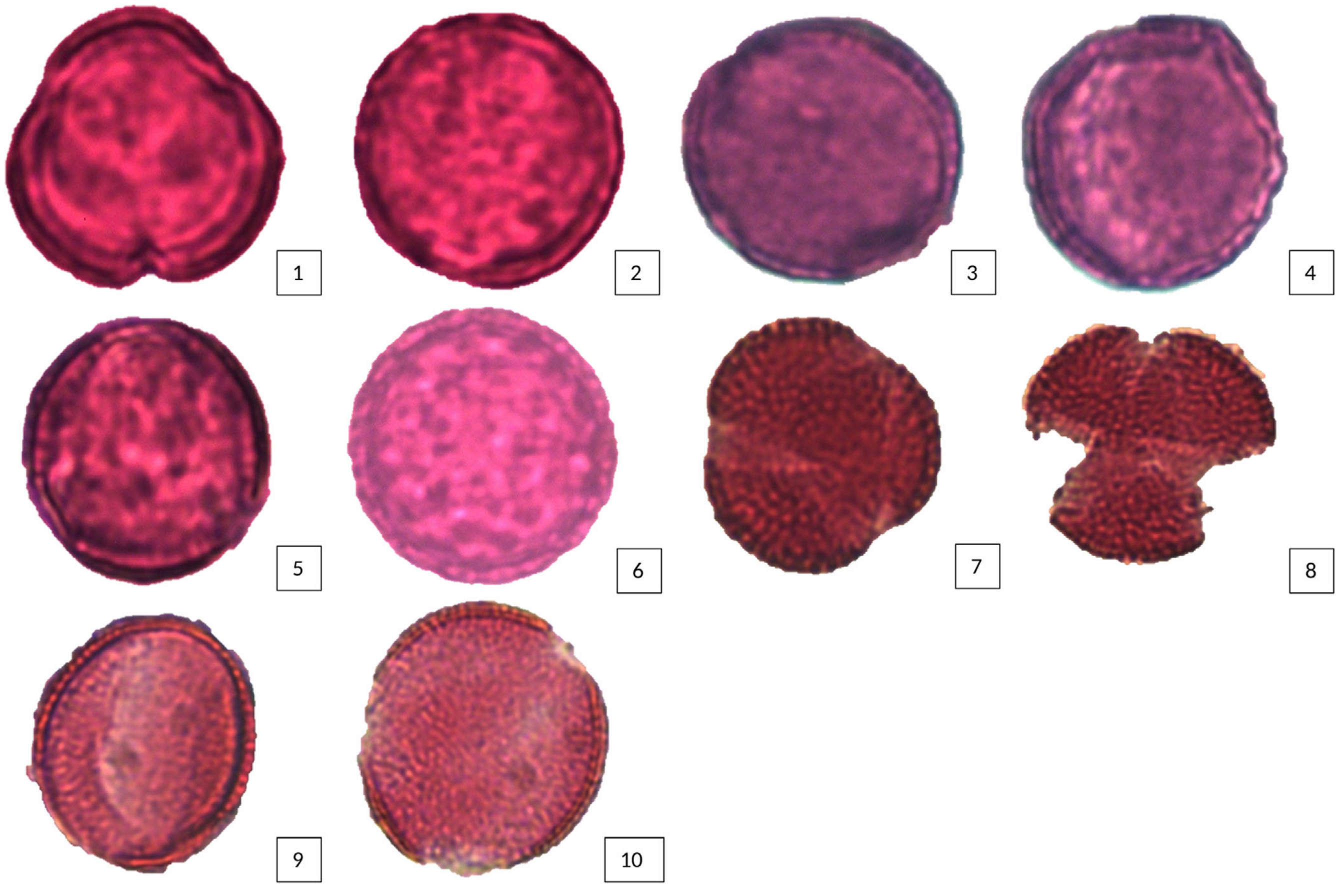

PLATE 1 Light micrographs of *Alliaria petiolate* (1 and 2), *Nasturtium officinale* (3), *Nasturtium microphyllum* (4), *Capsella bursa-pastoris* (5), *Lepidium didymium* (6), *Brassica rapa* (7 and 8), *Sisymbrium irio* (9 and 10) [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 2 Qualitative pollen characters of the taxa in family Brassicaceae

Name of taxa	Shape of pollen	Exine ornamentation	No. of apertures	Lumen shape
Alliaria petiolata (M.Bieb.) Cavara & Grande	Prolate	Reticulate	Tricolpate	Polygonal
Brassica rapes L.	Prolate	Reticulate	Tricolpate	Polygonal
Capsella bursa-pastoris (L.) Medik.	Prolate spheroidal to prolate	Micro-reticulate	Tricolpate	Circular to irregular
Cardamine hirsuta L.	Prolate	Coarsely reticulate	Tricolpate	Polygonal
Lepidium didymum L.	Prolate	Reticulate	Tricolpate	Polygonal
Nasturtium microphyllum Boenn. ex Reichenb	Prolate	Reticulate	Tricolpate	Polygonal
Nasturtium officinale W.T.Aiton	Sub-prolate	Reticulate	Tricolpate	Irregular
Rorippa indica (L.) Hiern	Prolate	Reticulate	Tricolpate	Irregular
Rorippa islandica (Oeder) Borbás	Prolate	Reticulate	Tricolpate	Irregular
Sisymbrium irio L.	Prolate	Reticulate	Tricolpate	Circular

TABLE 3 Quantitative characteristic of pollen of family Brassicaceae from Pakistan

Taxa	Polar diameter min (mean ± SE) max (μm)	Equatorial diameter min (mean ± SE) max (μm)	P/E ratio (μm)	Exine thickness min (mean ± SE) max (μm)	Length of colpi min (mean ± SE) max (μm)	Width of colpi min (mean ± SE) max (μm)	\Length of lumen (μm)	Width of lumen (μm)	Murus thickness (μm)
Alliaria petiolata (M. Bieb.) Cavara & Grande	20.6 (22.4 ± 0.8) 24.7	18.1 (21.2 ± 1.1) 23.5	1.05	0.9 (1.5 ± 0.3) 2.4	13.7 (15.4 ± 0.7) 17.1	1.8 (2.6 ± 0.2) 3.1	0.5 (1.4 ± 0.4) 2.3	0.1 (0.8 ± 0.3) 1.6	0.2 (0.5 ± 0.1) 0.6
Brassica rapes L.	23.6 (24.9 ± 0.5) 26.3	15.9 (20.8 ± 2.6) 27.7	1.20	1.3 (1.6 ± 0.1) 2	14.7 (19.7 ± 1.8) 23.1	1.3 (1.8 ± 0.3) 2.9	0.4 (1.3 ± 0.3) 1.9	0.1 (0.6 ± 0.1) 0.9	0.2 (0.4 ± 0.1) 0.9
Capsella bursa-pastoris (L.) Medik.	21.7 (24.1 ± 1.3) 27.2	16.9 (20.3 ± 2.0) 24.8	1.19	1.7 (2.02 ± 0.1) 2.3	14.9 (17.3 ± 1.05) 19.8	2.9 (3.3 ± 0.1) 3.7	0.3 (0.7 ± 0.1) 1.1	0.1 (0.4 ± 0.1) 0.8	0.12 (0.3 ± 0.1) 0.8
Cardamine hirsuta L.	32.0 (36.07 ± 1.77) 39.2	19.8 (21.50 ± 0.93) 23.7	1.68	1.2 (1.37 ± 0.10) 1.6	8.9 (9.52 ± 0.29) 10.2	1.9 (2.45 ± 0.23) 2.9	1.8 (2.10 ± 0.17) 2.6	0.9 (1.32 ± 0.20) 1.8	0.2 (0.30 ± 0.07) 0.51
Lepidium didymum L.	23.7 (26.5 ± 1.2) 29.3	12.6 (14.0 ± 0.5) 15.1	1.89	0.8 (1.5 ± 0.2) 2.1	7.2 (7.9 ± 0.2) 8.5	1.5 (1.9 ± 0.2) 2.8	1.7 (1.9 ± 0.1) 2.2	0.5 (1.05 ± 0.2) 1.5	0.1 (0.3 ± 0.08) 0.5
Nasturtium microphyllum Boenn. ex Reichenb	28.1 (29.6 ± 0.6) 31.1	17.5 (18.8 ± 0.5) 20.2	1.57	0.3 (0.7 ± 0.2) 1.5	6.9 (8.4 ± 0.5) 9.5	1.4 (2.0 ± 0.3) 2.9	1.4 (2.0 ± 0.2) 2.7	0.3 (0.8 ± 0.30) 1.7	0.23 (0.3 ± 0.05) 0.5
Nasturtium officinale W.T.Aiton	19.9 (22.3 ± 1.4) 26.1	17.9 (19.5 ± 1.0) 22.7	1.14	1.9 (2.2 ± 0.1) 2.7	16.1 (18.5 ± 0.8) 20.1	1.7 (2.1 ± 0.2) 2.9	0.4 (0.9 ± 0.2) 1.7	0.9 (1.1 ± 0.1) 1.4	0.1 (0.3 ± 0.1) 0.7
Rorippa indica (L.) Hiern	27.2 (30.07 ± 1.3) 33.2	14.5 (17.2 ± 1.1) 19.4	1.75	0.3 (0.6 ± 0.34) 1.1	15.5 (16.8 ± 1.22) 18.1	1.1 (1.7 ± 0.46) 2.2	0.7 (0.9 ± 0.16) 1.1	0.12 (0.6 ± 0.58) 1.4	0.1 (0.1 ± 0.04) 0.2
Rorippa islandica (Oeder) Borbás	32.8 (33.9 ± 0.6) 35.6	16.9 (18.5 ± 0.8) 20.1	1.83	0.2 (0.2 ± 0.04) 0.41	14.3 (16.6 ± 1.2) 19.4	2.1 (2.5 ± 0.2) 3.1	0.5 (1.05 ± 0.1) 1.4	0.3 (0.5 ± 0.1) 0.8	0.13 (0.4 ± 0.1) 0.6
Sisymbrium irio L.	25.3 (29.1 ± 1.9) 33.1	15.5 (16.9 ± 0.6) 18.6	1.72	0.1 (0.30 ± 0.07) 0.41	7.2 (9.2 ± 0.8) 11.3	3.2 (4.0 ± 0.4) 5.3	0.31 (0.97 ± 0.26) 1.6	0.1 (0.52 ± 0.1) 0.8	0.1 (0.30 ± 0.1) 0.6

2.3 | SEM analysis of pollens

Pollen of family Brassicaceae were prepared for SEM study using the techniques of Ullah et al. (2018) and Ahmad et al. (2018). For detail study of exine thickness and its ornamentation, SEM was used. For SEM, pollen were acetolyzed and then suspended in 90% ethanol. After this process, they were mounted on metallic stubs which were coated with gold palladium. Photographs of pollen taken using SEM (Model JEOL JSM-5910). The terminologies of Khalik Abdel et al. (2002) and Bahadur et al. (2018) were adopted for the ornamentation of exine of pollen.

2.4 | Statistical analysis

2.4.1 | Fertility and sterility percentage

Percentage of fertile and sterile pollen were calculated by using formula given by Ullah et al. (2018)

$$\text{Fertility} = \frac{F}{F+S} \times 100$$

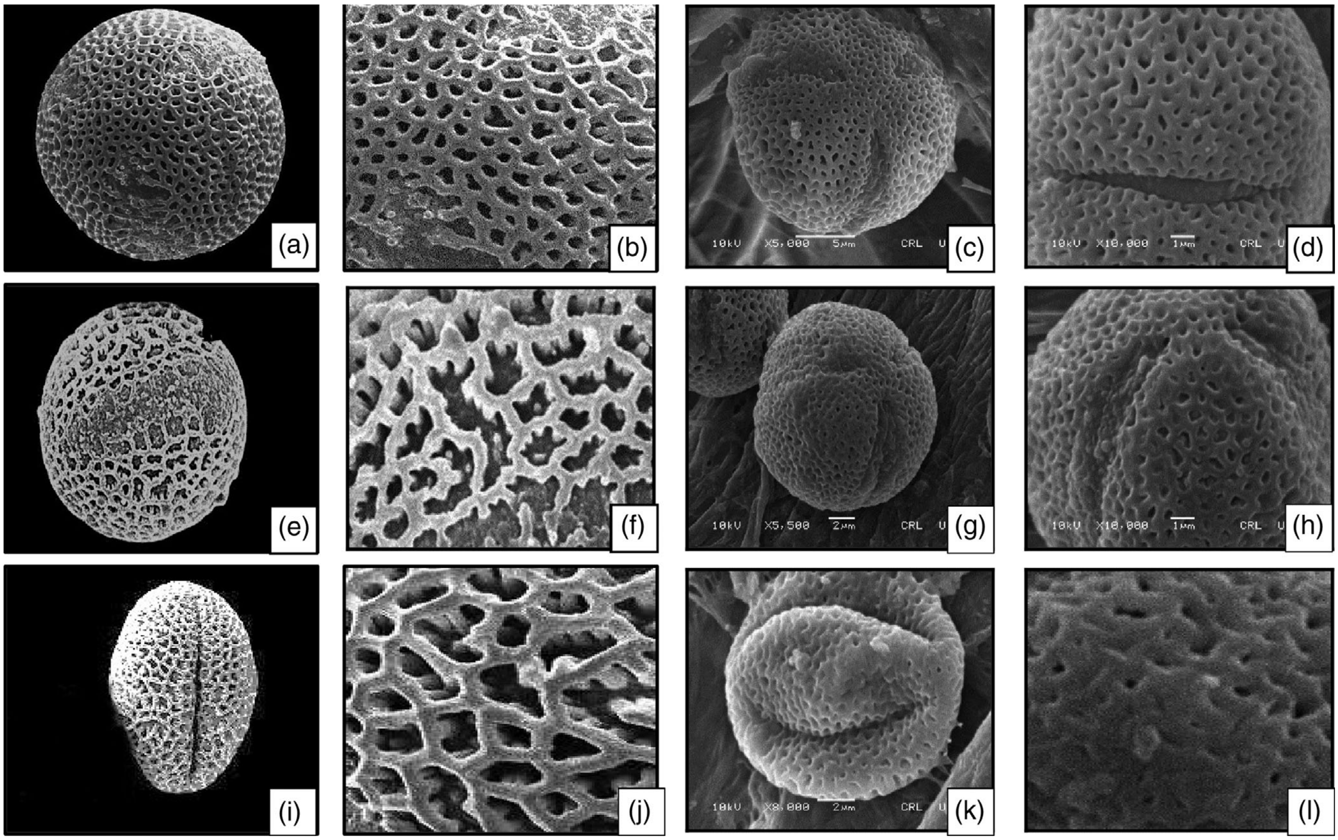

PLATE 2 SEM micrographs of *Capsella bursa-pastoris* (a and b), *Brassica rappa* (c and d), *Nasturtium officinale* (e and f), *Alliaria petiolata* (g and h), *Cardamine hirsute* (i and j), *Rorippa islandica* (k and l)

TABLE 4 Fertility and sterility percentage of pollens in family Brassicaceae

Sr. No.	Botanical name	No. of fertile pollen	No. of sterile pollen	Fertility (%)	Sterility (%)
1.	*Alliaria petiolata* (M.Bieb.) Cavara & Grande	73	18	80.22	19.78
2.	*Brassica rapa* L.	49	13	68.06	21.94
3.	*Capsella bursa-pastoris* (L.) Medik.	82	12	87.23	12.77
4.	*Cardamine hirsuta* L.	66	17	79.52	20.48
5.	*Lepidium didymum* L.	59	22	72.84	27.16
6.	*Nasturtium microphyllum* Boenn. ex Reichenb	61	19	76.25	23.75
7.	*Nasturtium officinale* W.T. Aiton	72	18	80.00	20.00
8.	*Rorippa indica* (L.) Hiern	48	9	84.21	15.79
9.	*Rorippa islandica* (Oeder) Borbás	69	22	75.82	24.18
10.	*Sisymbrium irio* L.	63	14	81.82	18.18

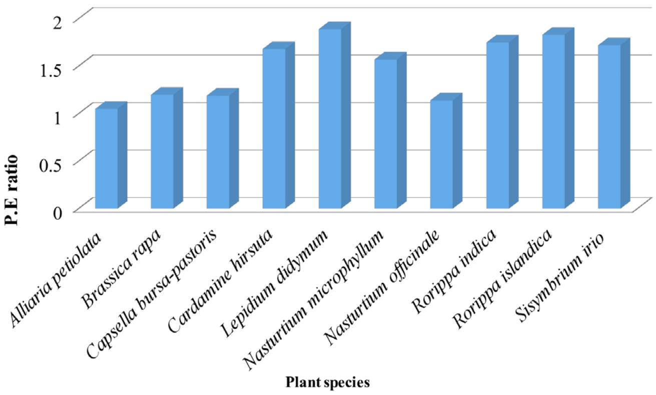

FIGURE 1 P.E ratio of various species of pollen of family Brassicaceae [Color figure can be viewed at wileyonlinelibrary.com]

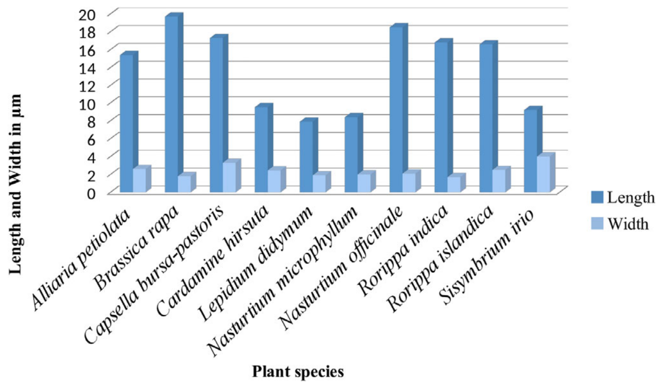

FIGURE 3 Comparison between length and width of Colpi in pollen grains [Color figure can be viewed at wileyonlinelibrary.com]

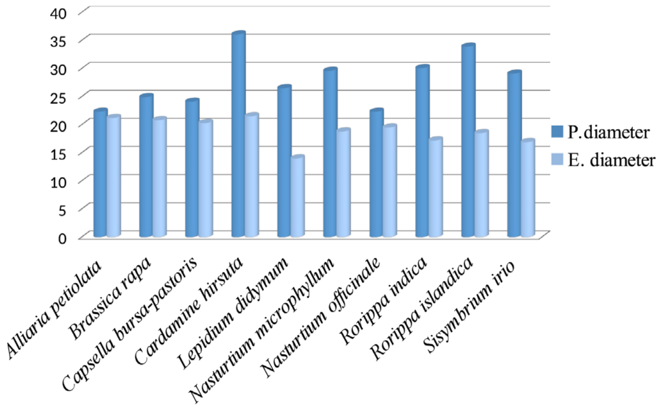

FIGURE 2 Comparison between polar and equatorial diameter of pollen in different species of family Brassicaceae [Color figure can be viewed at wileyonlinelibrary.com]

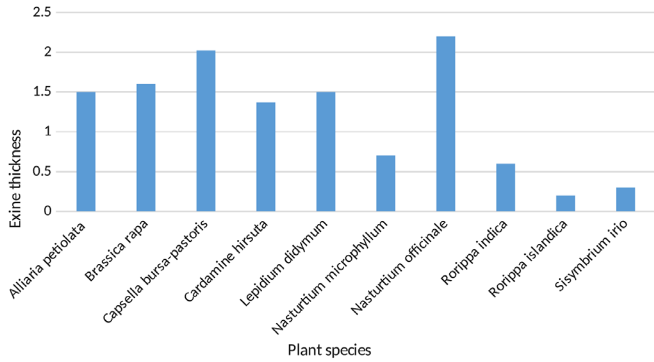

FIGURE 4 Variation in exine thickness of various species of family Brassicaceae [Color figure can be viewed at wileyonlinelibrary.com]

where F denotes the total number of fertile pollen and S denotes the number of sterile pollen

$$\text{Sterility} = \frac{S}{S+F} \times 100$$

2.4.2 | *P/E* ratio

This ratio were determined using the formula mentioned by Ahmad et al. (2018) in their study

$$P/E = \frac{P}{E}$$

where P is polar diameter and E is equatorial diameter of the same pollen.

3 | RESULTS

For the correct identification of members of same family or genus, pollen morphological characters such as pollen shape, size, exine sculpturing, the presence of colpi, pores, and lumen is of great importance. The pollen morphological characters both quantitative and qualitative obtained from LM and SEM are summarized in Tables 2 and 3. In present study, 19 species belongs to 7 genera of family Brassicaceae was examined (Plates 1 and 2). While studying, the pollen morphology slight variations were found in the shape of pollen as 8 species have same pollen shape prolate but *Nasturtium officinale* have sub-prolate and *Capsella bursa-pastoris* have spheroidal to prolate shape. All the pollen was tricolpate, and the ornamentation of exine was reticulate in all species except *Capsella bursa-pastoris* which have micro-reticulate sculpturing. Shape of lumen also varies among the species, five species have polygonal lumen shape, whereas *Sisymbrium irio* and *Capsella bursa-pastoris* have circular shape, and *Nasturtium officinale*, *Rorippa indica*, and *Rorippa islandica* have irregular shape of lumen. Sterility and fertility percentage of pollen were also calculated for each species. The highest number of fertility was found in *Capsella bursa-pastoris* (87.2%), whereas the lowest was observed in

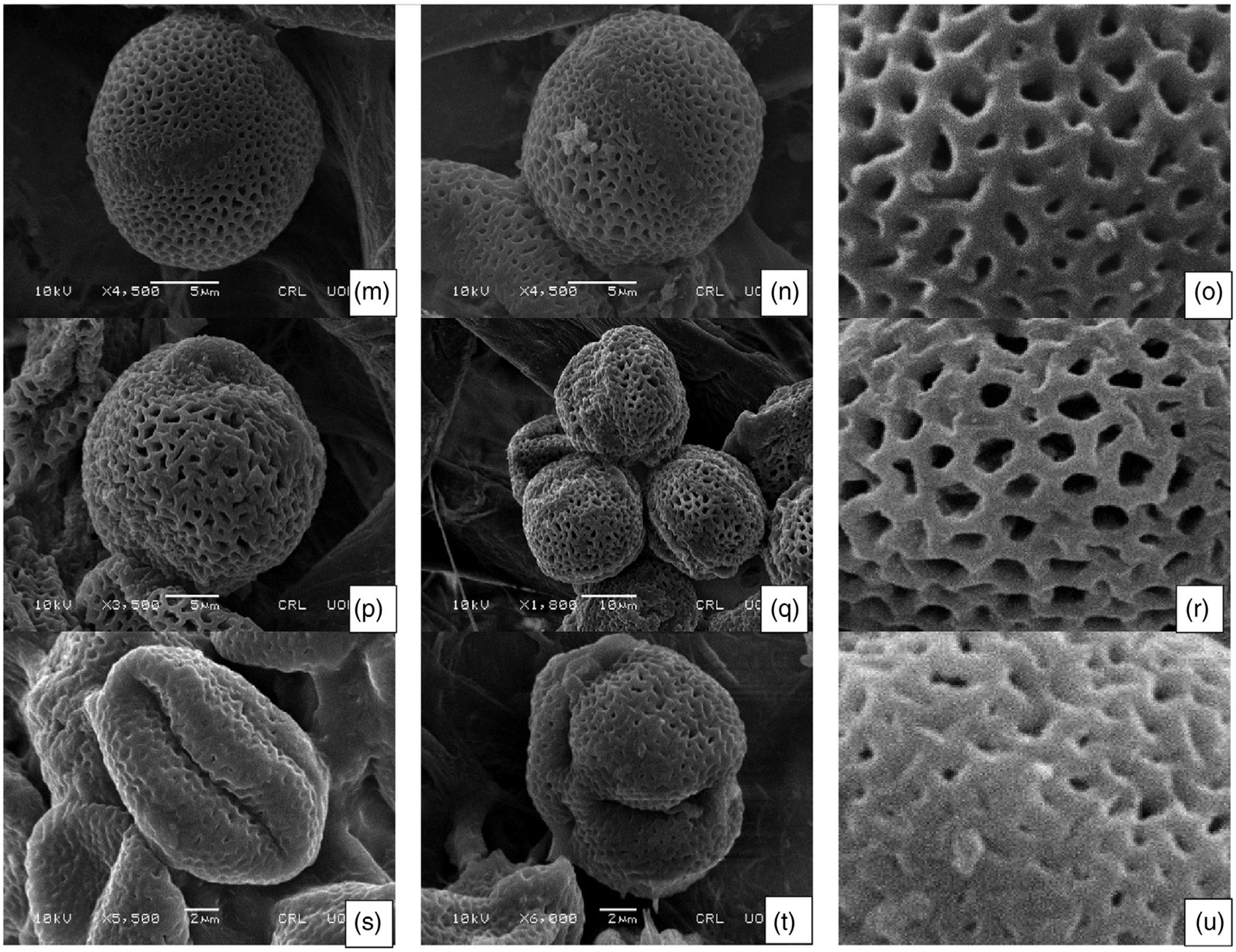

PLATE 3 SEM micrographs of *Lepidium didymium* (m–o), *Sisymbrium irio* (p–r), *Rorippa indica* (s–u)

Brassica rapa (68.06%). The maximum numbers of sterile pollen were found in *Rorippa islandica* (24.18%), and minimum sterile pollen were observed in *Capsella bursa-pastoris* (12.7%) (Table 4).

3.1 | Shape and size of pollens

The dominant shape of pollen in this study was prolate (eight species), whereas *Capsella bursa-pastoris* have spheroidal to prolate and *Nasturtium officinale* having sub-prolate pollen shape. The highest average diameter of pollen on polar axis was observed in *Cardamine hirsuta* (36.07 μm), whereas the lowest size was observed in *Nasturtium officinale* (22.3 μm). Similarly, maximum diameter of pollen in equatorial axis was observed in *Cardamine hirsuta* (21.5 μm), and minimum diameter was examined in *Lepidium didymum* (14.0 μm).

Size of the pollen was calculated on the basis of *P*/*E* ratio, among the species of Brassicaceae variations were found in size. The highest P.E value was observed in *Lepidium didymum* (1.89), and the lowest value was observed in *Alliaria petiolata* (1.05) (Figures 1 and 2).

3.2 | Micro-morphology of apertures

All the studied members of family Brassicaceae have tricolpate apertures. The size of apertures varies among the species as the maximum average length of colpi were observed in *Brassica rapa* (19.7 μm) and minimum length were found in *Lepidium didymum* (7.9 μm), whereas in terms of width, the maximum colpi was found in *Sisymbrium irio* L. (4.0 μm) and minimum width was observed in *Rorippa indica* (1.7 μm) (Figure 3).

3.3 | Exine ornamentation and murus thickness

Exine ornamentation is the most distinguishing character of the species (Plate 2). In this study, all the species have reticulate exine except *Capsella bursa-pastoris* (micro-echinate) and *Cardamine hirsuta* having coarsely reticulate exine sculpturing. Thickness of exine ranges from *Nasturtium officinale* (2.2 μm), whereas the minimum exine thickness was observed in *Rorippa islandica* (0.2 μm) (Figure 4). Murus is another character of pollen in family Brassicaceae. The size of murus thickness also varies among the species as it ranges from *Alliaria petiolata* (0.5 μm) maximum to *Rorippa indica* (0.1 μm) with minimum murus thickness (Plate 3).

3.4 | Morphological characters of lumen

Lumen formation is the characteristic of reticulate exine ornamentation. In the studied species of family Brassicaceae, shape of lumen is polygonal except *Capsella bursa-pastoris*, *Sisymbrium irio* with circular shape and *Nasturtium officinale*, *Rorippa indica*, *Rorippa islandica* have irregular shape of the lumen. Size of the lumen also varies among the species as the highest length of lumen was found to be *Cardamine hirsuta* (2.10 μm) and width was observed in *Cardamine hirsuta* (1.32 μm).

3.5 | Taxonomic key

1.	+ Shape of pollen prolate and exine ornamentation reticulate……………*Alliaria petiolata*
	– Shape of pollen prolate spheroidal and exine ornamentation micro-reticulate……….(3)
2.	+ Lumen shape circular and exine ornamentation micro-reticulate…*Capsella bursa-pastoris*
	– Lumen shape polygonal and exine ornamentation coarsely reticulate……………….(4)
3.	+ Exine ornamentation coarsely reticulate and lumen shape polygonal……*Cardamine hirsuta*
	– Exine ornamentation reticulate and lumen shape polygonal………………………….(5)
4.	+ Shape of pollen sub-prolate and lumen shape irregular………………*Nasturtium officinale*
	– Shape of pollen prolate and lumen shape irregular…………………………………….(8)
5.	+ Exine ornamentation reticulate and lumen shape irregular……………… *Rorippa islandica*
	– Exine ornamentation reticulate and lumen shape circular …………………………….(10)

4 | DISCUSSION

In present research, pollen morphology of different species of family Brassicaceae was studied using LM and SEM. By using these pollen morphological characters, taxonomic keys have been constructed for easy identification and delimitation of taxa. All the species mentioned in this research have tricolpate type of pollen with reticulate exine ornamentation. The shape of pollen was prolate and sub-prolate, and lumen shape differs among the species. Perveen et al. (2004) reported the pollen morphology of several species of family Brassicaceae having reticulated exine ornamentation. Appel and Al-Shehbaz (2003) worked on pollen morphology of family Brassicaceae and observed all pollen grains having tricolpate reticulate type. Moore, Webb, and Collison (1991) classified the family in tricolpate with reticulate surface of exine. Previously some work has been done on the palynology of family Brassicaceae by Brochmann (1992) and Al-shehbaz, Beilstein, and Kellogg (2006). They believed that shape of lumen is important character in identification of different species of family Brassicaceae. In this study, some species have same shape of lumen as *Nasturtium officinale* and *Rorippa indica* have same lumen shape but they differ from each other through size of pollen. Based on lumen shape, exine ornamentation is divided into three types reticulate, micro-reticulate, and coarsely reticulate. Studies conducted by Anchev and Deneva (1997) reported two types of exine, that is, reticulate sculpturing with lumina equal in size to the muri and perforated reticulate sculpturing with lumina smaller than the muri.

Apart from exine ornamentation, thickness of exine is also important character to distinguish species of this family. Erdtman (1986) in his study divided the species of Brassicaceae based on exine thickness into two pollen types. According to Albermani, Albermani, and Altameme (2017), thickness of *Nasturtium officinale* is 1–2 μm which is different from the results reported in this study, difference is due to climatic conditions of the area. Sub-prolate pollens of *Nasturtium officinale* have been observed in this study which resembles with the study conducted by Perveen et al. (2004), whereas Lahham and Al-Eisawi (1987) reported sub-spheroidal pollen type in *Nasturtium officinale*.

In this study, the shape of pollen of *Cardamine hirsuta* and *Lepidium didymum* is prolate having polygonal lumen shape, and this result was similar to previous study reported by Gabr (2018). Exine thickness of *Cardamine hirsuta* is 1.37 μm observed in current study which disagrees with the study conducted by Khalik Abdel et al. (2002) and Appel and Al-Shehbaz (2003). Anchev and Deneva (1997) studied palynology of family Brassicaceae, reported some characters to differentiate species of the family, that is, shape of pollen, exine sculpturing, number of apertures, and the pattern of lumina. According to our findings, these characters are not sufficient for classification of members of the family; sometimes, the pollen of different species had similar characteristics. The most important characters in the present study were lumen length, width, murus thickness, and exine thickness.

5 | CONCLUSION

In present work, morphology of pollen offers some significant structures that aid in the delimitation of different species in family Brassicaceae. Investigation through SEM revealed pollen morphological characters which are not seen through light microscope. Surface of pollen, its shape, and wall sculpturing demonstrated to be valuable features at generic as well as specific level within the family. In present findings, pollen sculpturing, shape, and exine thickness shows variations which are the basic characters to identify pollen of the Brassicaceae species. These characters play crucial role in the correct identification of plants and delimitation of complex species. It is therefore recommended to study pollen morphological features to identify taxonomically problematic taxa.

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CONFLICT OF INTEREST

All authors declare that they have no conflict of interest.

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